1a) r(t) = (t,t^3) 0<=t<=2
integral(0-2)(t^2+2t^6)*(sqrt(1+(3t^2)^2))dt
b) integral(0-2)(t^2+2t^6)*3t^2dt
integral(0-2)(3t^4+6t^8)dt
3/5t^5+6/9t^9 from(0-2)
3/5(2)^5+6/9(2)^9
c)integral(0-2)(t^2+2t^6)*1dt
t^3/3+2/7t^7 from(0-2)
8/3+2/7(2)^7

Worksheet 10, 16.1-2

2
integral(0-2)(t^2+2t^6+t^2+2t^6)dt
integral(0-2)(2t^2+4t^6)dt
2/3t^3+4/7t^7 from(0-2)
16/3+4/7*2^7
1b is the dot product of (x(t),y(t)) with (0,dy) and 1c is the dot product of (x(t),y(t)) with (dx,0) in question 2 we simply take the dot product of (x(t),y(t)) with (dx,dy) which is the same as adding 1b and 1c

Practice Problems

1. Let $f(x, y) = x^2 + 2y^2$ and C be the curve along $y = x^3$ from $(0, 0)$ to $(2, 8)$. Find each of the following:

   (a) $\int_C f(x, y)\, ds$ (Setup but do NOT evaluate)

   (b) $\int_C f(x, y)\, dy$ (Evaluate, should get 5408/15)

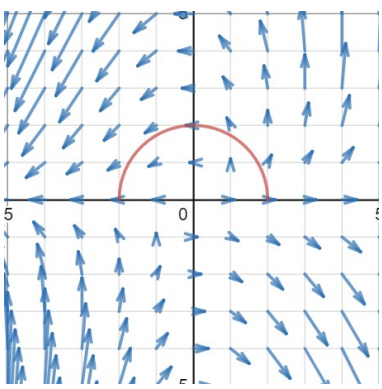
   (c) $\int_C f(x, y)\, dx$ (Evaluate, should get 824/21)

2. Let $\vec{F} = \langle x^2 + 2y^2, x^2 + 2y^2 \rangle$ and C be the same curve as in number 1. Find

$$\int_C \vec{F} \cdot d\vec{r}.$$

   Then show this is the same as 1b+1c and explain why.

3. Let $\vec{F} = \langle x - y, xy \rangle$ and C be arc of the circle $x^2 + y^2 = 4$ traversed counter-clockwise from $(2, 0)$ to $(-2, 0)$. A sketch of the vector field a curve are below.



   (a) Using the image above, determine whether

$$\int_C \vec{F} \cdot d\vec{r}$$

   positive

   is positive or negative.

   (b) Evaluate

$$\int_C \vec{F} \cdot d\vec{r}.$$

   b
   x = 2cos(t)
   y = 2sin(t)
   r(t) = <2cos(t),2sin(t)>
   0<=t<=pi
   dr = <-2sin(t),2cos(t)>
   F = <2cos(t)-2sin(t),4cos(t)sin(t)>
   integral(0-pi)-4cos(t)sin(t)+4sin^2(t)+8cos^2(t)sin(t)
   2pi+16/3

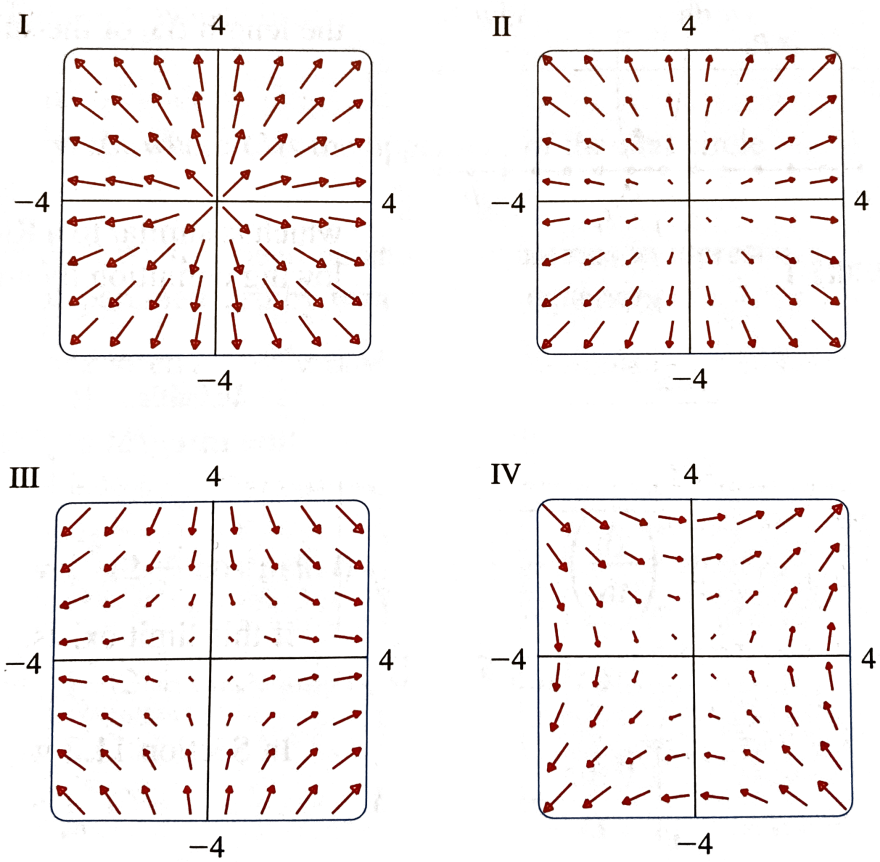
   You may use that $\int_0^\pi \sin^2(t) = \pi/2$. (Hint: the answer is $16/3 + 2\pi$)

Extra Quiz/Exam Practice

4. Let $\vec{F}(x,y) = \langle y, x\rangle$ and let C be the line segment starting at (2,0) and going to (3,3). Evaluate the line integral
$$\int_C \vec{F}\cdot d\vec{r}$$

5. Let $f(x,y) = x + y$. Find
$$\int_C f(x,y)ds$$
for each C below.
   (a) C is the curve from $(1,0)$ to $(0,1)$ along the a straight line path
   (b) C is the curve from $(0,1)$ to $(1,0)$ along the a straight line path
   (c) Note that these are the same values.

6. Match each of the functions below with the plots of the gradient vector fields. Explain your reasoning.



$$f(x,y) = xy \qquad f(x,y) = x^2 + y^2 \qquad f(x,y) = x^2 - y^2 \qquad f(x,y) = \sqrt{x^2+y^2}$$

7. Find the work done by the force field $\vec{F}(x,y) = \langle x\sin(y), y\rangle$ on a particle that moves along the parabola $y = x^2$ from $(-1,1)$ to $(2,4)$. (Hint: ans ≈ 8.1).

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Challenge Questions

8. Recall that C (from number 1) is a curve in the xy plane. We can get a curve in $\mathbb{R}^3$ by letting the z-coordinate be given by $f(x, y)$. Call this new curve C'. We've seen that $\int_C f(x, y)\, ds$ gives the area under the curve C' in $\mathbb{R}^3$. To visualize this, see `https://www.geogebra.org/3d/us8v9ggs` (blue area). We've also seen $\int_C f(x, y)\, dy$ gives the area under the curve when it is projected into the yz plane (green area). Let's find that green area another way.

   (a) First, let's find the curve. One way of writing C' is $\langle t, t^2, t^2 + t^4 \rangle$. Explain why this works.

   (b) Now, we need to project C' into the yz-plane. We do this by changing the x coordinate in its parametrization to 0. Do this. Call the new curve C''. (This part's easy, literally just change the first component to 0 and keep the rest the same.)

   (c) Now let $y = t^2$ and get an expression of the form $z = g(y)$ for the curve C'' in the yz-plane.

   (d) Use single variable calculus techniques to find the area under the curve C''. This should be the same value as 1b.

9. Let $\vec{F}$ be a vector field and the curve C be the path along the invertible function $y = f(x)$ from (a, c) to (b, d). Two possible parameterizations of the curve C are $\langle t, f(t) \rangle, a \leq t \leq b$ and $\langle f^{-1}(s), s \rangle, c \leq s \leq d$. Show that we get the same values for $\int_C \vec{F} \cdot d\vec{r}$ independent of the choice between these two parameterizations.

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